Jonathan Andujo

Principal-Level Software Engineer | AI Platform & Cloud Infrastructure

Microsoft Copilot Tuning | Azure AI | LLM Fine-Tuning | .NET

jonathan_andujo@outlook.com +52 614 217 1150 linkedin.com/in/jandujo github.com/jonathanandujo

EXECUTIVE SUMMARY

Principal-level software engineer with 19+ years of experience architecting and delivering enterprise platforms, cloud services, and commercial software. At Microsoft, contribute across Copilot Tuning and Azure AI infrastructure, combining hands-on engineering with platform architecture, cross-team enablement, and technical leadership. Designed an internal engineering platform adopted by 400+ engineers and delivered systems and optimizations associated with 800K+ requests per second, 10x traffic capacity, and substantial reductions in cost and latency. Deep experience in Azure Machine Learning, Azure OpenAI, SFT/RFT workflows, .NET, distributed systems, and full-stack development.

CORE EXPERTISE

AI platforms: Microsoft Copilot Tuning, Azure Machine Learning (AML), Azure OpenAI (AOAI), supervised fine-tuning (SFT), reinforcement fine-tuning (RFT), evaluation workflows, responsible AI

Cloud and architecture: Microsoft Azure, AWS, infrastructure as code (IaC), Bicep, Kubernetes, Azure Data Factory, Azure Data Lake Storage Gen2, Cosmos DB, distributed systems, high availability, cost optimization

Engineering: C#, .NET 10, .NET Core, .NET Framework, SQL Server, React, Angular, JavaScript, TypeScript, Node.js, REST APIs, Redis, CI/CD, Azure Pipelines, Git, YAML

Quality and leadership: NUnit, Moq, SpecFlow, Microsoft Playwright, SonarQube, software architecture, technical strategy, mentoring, Agile, SOLID

PROFESSIONAL EXPERIENCE

Senior Software Engineer, Copilot Tuning (Principal-level platform scope) — **July 2025–Present**

Microsoft | Remote

- Drive Copilot Tuning platform engineering, with a focus on model fine-tuning, GPU efficiency, and resource utilization to improve scalability and reduce operating costs.
- Designed and built an internal engineering platform adopted by 400+ software engineers, replacing 5+ disconnected tools and consolidating workflows for Copilot Tuning projects, Azure Machine Learning jobs, Azure OpenAI capacity, and cross-team dependencies.
- Architected self-service Azure Machine Learning infrastructure provisioning and lifecycle automation, including workspaces, compute, managed identities, private endpoints, data stores, and scheduled jobs; reduced environment setup from days to minutes.
- Successfully upgraded 50+ projects from .NET 8 to .NET 10, ensuring platform compatibility, modernizing dependencies, and minimizing migration risks.
- Built Azure OpenAI account automation covering onboarding, quota and capacity management, provisioned throughput allocation, regional failover, monitoring, and role-based access control, enabling product teams to onboard independently.
- Designed orchestration for SFT and RFT workflows across Power Automate, Azure Data Factory, and Azure Machine Learning, supporting job dispatch, dataset validation, responsible AI checks, evaluation, and model publishing.
- Developed scale-testing capabilities for AI platform dependencies; a related .NET Framework-to-.NET 9 migration and end-to-end load-testing effort reached 800,000 requests per second.

Senior Software Engineer, IC3 Customer Insights — **April 2023–July 2025**

Microsoft | Remote

- Reduced build pipeline execution time by 67% by migrating to a build engine using parallel processing and caching.
- Migrated APIs from .NET Framework to .NET Core to improve performance, reliability, and cross-platform support.
- Deployed Kubernetes services for Microsoft Cosmic API in APAC, improving scalability, fault tolerance, and load balancing.
- Mentored engineers in SQL Server query design, indexes, statistics, and database engineering practices.

Senior Software Engineer, IC3 Data Platform (Neuron) — **August 2021–April 2023**

Microsoft | Remote

- Built an internal project-management portal adopted by more than 10,000 concurrent users, bringing together multiple data sources with React and .NET Core.
- Reduced SQL query response time from 2 seconds to 150 milliseconds, enabling 10x traffic capacity on existing infrastructure.
- Led a migration from Node.js to React and .NET Core and implemented end-to-end integration tests with SpecFlow and Microsoft Playwright.
- Mentored three interns building a React and .NET Core platform to manage thousands of Azure Data Factory pipelines.
- Worked with large-scale data platforms using Azure Data Factory and Azure Data Lake Storage Gen2.

Lead Software Engineer **November 2020–August 2021**

Civix | New Orleans, Louisiana

- Automated continuous integration pipelines with Git, Azure Pipelines, and YAML.
- Reduced SQL Server stored procedure execution time by 90% through query and execution-plan optimization.
- Mentored developers in C#, .NET, SOLID principles, and clean architecture.

Software Engineer Lead / .NET Architect **January 2019–October 2020**

Tiempo Development | Guadalajara, Mexico

- Designed and led development of Azure-hosted solutions for more than 3,000 users using .NET Core 3.1, Angular, and Cosmos DB.
- Designed application architecture and Azure infrastructure from the ground up; optimized resource use to reduce costs by 70%.
- Reduced defects by 21% by strengthening unit testing with NUnit and Moq and promoting SonarQube-validated engineering practices.

Software Engineer Lead / Architect **April 2018–January 2019**

Grupo Ultra | Cancún, Mexico

- Delivered software for sales, warehouse, and support operations using SQL and .NET Core; increased sales revenue by 15% through improved business rules.
- Built a warehouse label-printing solution that avoided rework and saved $10,000 in operating costs.
- Reduced sales commission calculation time by 85% with configurable rules built on .NET Core 2.1 and Angular.
- Implemented a help-desk tool that improved customer satisfaction by 20% and modernized a core system using AWS services.

Senior Software Engineer **October 2016–April 2018**

Best Day Travel Group | Cancún, Mexico

- Designed and implemented travel-platform architecture supporting more than 150,000 requests per second.
- Introduced Redis caching to improve portal performance by 50%; used multithreaded queries across more than 1,000 sources to reduce response time by 30%.
- Integrated supplier pricing and business rules that reduced booking costs by an average of 15%.

Software Engineer Lead **June 2014–September 2016**

TAC Global Solutions | Mexico

- Led development of ERP and CRM applications using VB.NET, SQL Server, WCF, and AngularJS; helped build a mobile warehouse-management application for inventory and pallet tracking.
- Improved CRM lead conversion by 15% and shortened the sales cycle by 10% through sales-funnel KPIs; used geolocation analytics to increase regional marketing ROI by 5%.
- Reduced billing discrepancies by 70%; improved inventory planning to cut overstock by 20% and save 12% in costs.

Software Developer **July 2012–April 2014**

ADS Systems | Mexico

- Developed an RFID telemetry payment system and contributed to a secure, scalable ERP architecture.
- Reduced SQL Server warehouse picking time by 90% by replacing cursor-based operations with common table expressions; cut handheld inventory processing from 2.5 hours to 15 minutes per day.
- Built a Mexico-compliant payroll system in under four weeks, integrating external tax-validation services.

Web Developer **July 2011–July 2012**

Grupo Industrial Advantage Tecnologia | Mexico

- Developed an e-commerce shopping cart using JavaScript and PHP and improved website SEO through keyword, metadata, and content optimization.
- Maintained a Visual Basic 6 school-ring application and developed a VB.NET queue-management application for banks and government offices.

Software Developer and Architect, Freelance **September 2007–October 2020**

VYTES | Mexico

- Designed and delivered business software and electronic invoicing products using .NET Framework and SQL Server; the invoicing service supported more than 40 clients on infrastructure costing under $50 per month.
- Optimized a payroll system to process more than 60,000 requests in 25 minutes on a development-tier server, improving performance by 80%.
- Applied user-centered design and agile practices to deliver maintainable software tailored to client needs.

EDUCATION

Master's in IT Management — TecMilenio University

Bachelor's in IT Engineering — Universidad Tecnológica de Chihuahua

CERTIFICATIONS

Azure Fundamentals (AZ-900) — Microsoft

Azure AI Fundamentals (AI-900) — Microsoft

Cloud Practitioner Essentials — AWS

Credential profile: credly.com/users/jonathan-andujo-flores/badges